Case-Study Parameters

1 Modeling and Cost Settings

Tables 1 and 2 summarize the resource, regulation, security, and cost settings. The EV and TCL fleets are represented by aggregate virtual batteries. Communication latencies follow wireless-network delay characteristics, with subsequent resource responses modeled by first-order dynamics. The dynamic model uses per-unit power and frequency deviations on 100-MW and 50-Hz bases.

All cost functions share the quadratic form derived from [1]:

$$\phi(x;\theta) = \theta_1 x + \theta_2 x^2, \qquad \theta = (\theta_1, \theta_2), \tag{1}$$

$$C_i^{\mathrm{inc}}(u) = \Delta t\, \phi(u; \theta_i^{\mathrm{inc}}), \tag{2}$$

$$C_i^{\mathrm{reg}}(r) = \phi(r; \theta_i^{\mathrm{reg}}), \tag{3}$$

$$C^{\mathrm{sup}}(H, D) = \phi(H; \theta_H) + \phi(D; \theta_D). \tag{4}$$

Here, $u = U_i^{\mathrm{inc}}(t)$ and $r = U_i^{\mathrm{reg}}(t)$ are expressed in MW, and Δt in hours. Load-support costs represent resource bids; reserve costs represent ESS degradation, EV departure-SOC deviation, and TCL comfort deviation. The support cost is charged once over the maintenance window. Coefficients are listed in Table 2.

2 Resource Operating Parameters

Table 1: Local System Resource Operating Parameters

Parameter	Value
SG / GFM scheduled output	90 MW / 10 MW
ESS power / energy	45 MW / 100 MWh
ESS SOC	Initial: 75%; bounds: 20–100%
ESS efficiency	95% (charging and discharging)
EV battery / power	50 kWh / 7 kW
EV arrival / departure [2]	$\mathcal{N}(8.43, 1.12^2)$ / $\mathcal{N}(18.51, 1.03^2)$ h
EV SOC	Connection: $\mathcal{U}(0.2, 0.6)$ [2]; required departure: 0.90; bounds: 0.10–1.00
TCL unit rating / COP	2.5 kW / 3.5
TCL comfort band	$23.9 \pm 1.8\,^{\circ}\mathrm{C}$
TCL thermal parameters	C^{th}, R^{th} from [3]

3 Regulation, Security, and Cost Parameters

Table 2: Regulation, Security, and Cost Parameters

Resource parameter	ESS	EV	TCL
Mean latency (ms)	372.9	555.2	801.2
Time constant (ms)	50	80	100
$\alpha_i^{\min}$	0.09	0.08	0.06
$\alpha_i^{\max}$	0.90	0.80	0.60
Load support θ_i^{inc}	$(0.950, 0.018)$	$(1.050, 0.015)$	$(1.121, 0.014)$
Reserve θ_i^{reg}	$(0.020, 0.114)$	$(0.018, 0.097)$	$(0.025, 0.164)$

System parameter	Value
Base power / frequency	100 MW / 50 Hz
Frequency-deviation limit	0.3 Hz
RoCoF limit	1 Hz/s
GFM dynamic headroom	5 MW
Inertia bounds	$H \in [0.5, 1.8]$ s
Damping bounds	$D \in [1.5, 6]$
Support cost θ_H	$(60, 1.2)$
Support cost θ_D	$(10, 0.2)$

Note: Reported latencies are mean values; simulation latencies are sampled from the distributions reported in [4].

References

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